

# STAR Project

## Combined Reference

Spatial Topography Accessibility Robot

Version 1.1

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MIT License

STAR Development Team

Locked Inc.

*The everything-PDF: engineering reference, every per-module firmware Doxygen PDF (src, tests, 27 libs, 11 bench apps), gateway / ROS2 / proto / compliance / MATLAB source listings, the production Grafana dashboard + Pi 5 ops surface, capstone deliverables (SDD, SSUM, Test Report, Demo, Presentation), and the mechanical drawings + electrical schematic / PCB package – bundled into a single PDF for offline distribution.*

*Each section below is its own self-contained PDF with its own internal table of contents and bookmarks; navigate via the PDF viewer's bookmark tree.*

## Section overview

- Part I – Engineering Reference** Comprehensive technical book: hardware pinout, communication stack, NASA Power-of-10 rules, gateway architecture, ROS2 integration, motor control loop, safety architecture, test strategy, troubleshooting, glossary, compliance engine, operations runbook, logging architecture (`docs/star_documentation.pdf`, ~241 pages).
- Part II – Firmware Combined PDF** Every per-module Doxygen PDF for the RX72N firmware: `src/` (application), `tests/` (host unit-test suite), all 27 libraries under `libs/`, and all 11 bench-test applications. ~16,000 pages, 128 MB (`docs/STAR_Firmware.Combined.pdf`).
- Part III – Gateway Combined PDF** Go gateway: `cmd/` (virtual\_rx72n + cli tools), `internal/` (manager / controller / service / transport / WebSocket), and `tests/` as syntax-highlighted source listings (469 pages).
- Part IV – ROS2 Combined PDF** All 9 STAR-owned ROS2 packages as syntax-highlighted source listings: `star_bringup`, `star_compliance_msgs`, `star_gateway_bridge`, `star_perception`, `star_safety_monitor`, `star_simple_bridge`, `star_simulation`, `star_spi_bridge`, `star_supervisor` (~226 pages).
- Part V – Protocol Buffers Combined PDF** All proto3 schemas under `star-proto/proto/star/v1/` (33 pages).
- Part VI – Compliance Combined PDF** Python ADA-auditing engine under `star-compliance/` (~85 pages).
- Part VII – MATLAB Sources** System identification + PID design scripts under `matlab/` (39 pages).
- Part VIII – Operations** Production Grafana dashboard JSON + rendered panel screenshots + Pi 5 cockpit-API server + systemd units (`monitoring/` + `infra/pi5/`, 83 pages).
- Part IX – Software Description Document** Capstone deliverable.
- Part X – System and Software User Manual** Capstone deliverable.
- Part XI – Test Report** Capstone deliverable.
- Part XII – Final Demo Handout** Capstone deliverable.
- Part XIII – Final Presentation** Capstone deliverable (PowerPoint export).
- Part XIV – Mechanical Drawings** Master + 9 individual parts (Battery Box, Drivetrain Assembly, Lid L, Lid R, Motor mount, Sheet base, Sheet Lidar, Side panel, Split rail).
- Part XV – Electrical Schematics + PCB** KiCad outputs: Electrical Design + Schematic + PCB layout.
- Part XVI – Miscellaneous** Bench schematic, camera calibration, TOM board build instructions, compliance audit sample.